MIL-W-63181(AR)
17 November 1977
SUPERSEDING
See Section 6

# MILITARY SPECIFICATION

## WARHEAD SECTION SUBASSEMBLY, ATOMIC: M15

This specification is approved for use by the US Army Armament Research and Development Command and is available for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope.- This specification covers one type of warhead section subassembly; designated Warhead Section Subassembly, Atomic: M15; which is a part of the Pershing warhead section.

## 2. APPLICABLE DOCUMENTS

2.1 Issues of documents.- The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-S-14767 - Spin System, Atomic Warhead Section: M30.
MIL-S-45894 - Switch, Sensitive: M22.
MIL-F-48003 - Fuze, Impact, Atomic Weapon: M1131.
MIL-A-48316 - Amplifier, Trigger, Pulse: M4E1.
MIL-F-50279 - Fuze, Inertial: M1103.

**Beneficial comments (recommendations, additions, deletions), and any pertinent data which may be of use in improving this document, should be addressed to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QAN-Q, Dover, New Jersey 07801, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426), appearing at the end of this document, or by letter.**

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MIL-B-50897 - Battery, Thermal: M8.
MIL-S-50951 - Switch, Rotary: M21A1.
MIL-B-50957 - Battery, Thermal: M9.
MIL-C-50963 - Cables, Warhead Section.
MIL-F-50968 - Fuze, Impact, Atomic Weapon: T53E2.

STANDARDS

*MILITARY*

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

DRAWINGS (see 6.5)

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

8806652 - Marking and Sealing for Cylindrical Containers.
8876120 - Warhead Section Subassembly, Atomic: M15.
8886529 - Shipping and Storage Container: M483, Packing and Marking, for Warhead Section Subassemblies, Package 2 of 2.
8886530 - Shipping and Storage Container: M483, Packing and Marking, for Warhead Section Subassemblies, Package 1 of 2.
10560378 - Aft Section, Missile Warhead Section, Structure.
10560388 - Center Section, Structure.
10560543 - Nose Section, Missile Warhead Section, Structure.
10561209 - Ballistic Case Assembly.
10595610 - Ring, Splice (Aft Warhead).

PUBLICATIONS (see 6.5)

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

QAI-PA-1 - Quality Assurance Instructions for Reporting Data on Nuclear Weapons.

Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.



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3. REQUIREMENTS

3.1 Materials and components.- Materials shall conform to applicable specifications and drawings. Each component shall conform to all dimensions and tests specified on the component drawing. The contractor shall have available verifiable proof (i.e., objective evidence (see 6.2)) that materials and components were fabricated, inspected, and tested under controllable conditions as set forth in the contractor's quality control or inspection procedures.

3.2 Assembly.- The M15 warhead section subassembly shall comply with all requirements specified on drawing (dwg.) 8876120, with all requirements specified herein, and with the requirements of applicable specifications to the extent specified on the drawings and herein.

3.3 Components.

3.3.1 Nose section.- The nose section shall be procured and shall comply with dwg. 10560543.

3.3.2 Center and aft sections.- The center and aft sections shall be procured and shall comply with dwg. 10560388 and 10560378 respectively.

3.3.3 Ballistic case assembly.- The ballistic case assembly shall be procured and shall comply with dwg. 10561209.

3.3.4 M30 spin system.- The M30 spin system shall be procured and shall comply with MIL-S-14767.

3.3.5 Warhead section cables.- The warhead section cables shall be procured and shall comply with MIL-C-50963.

3.3.6 Fuzes.- The M1103, T53E2, and M1131 fuzes shall be procured and shall comply with MIL-F-50279, MIL-F-50968, and MIL-F-48003 respectively.

3.3.7 M4E1 trigger pulse amplifier.- The M4E1 trigger pulse amplifier shall be procured and shall comply with MIL-A-48316.

3.3.8 M21A1 rotary switch.- The M21A1 rotary switch shall be procured and shall comply with MIL-S-50951.

3.3.9 M22 sensitive switch.- The M22 sensitive switch shall be procured and shall comply with MIL-S-45894.

3.3.10 Batteries.- The M8 and M9 thermal batteries shall be procured and shall comply with MIL-B-50897 and MIL-B-50957 respectively.



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3.4 First article.- When specified, the contractor shall submit sample warhead section subassemblies for first acticle inspection and approval (see 4.3 and 6.1.1).

3.5 Capacitance of the impact fuze networks.- The capacitance of the impact fuze networks shall be as specified in Table I. The tests on the nose section networks shall be performed prior to the tests on the overall networks. In addition, the actual capacitance value in picofarads (pf) for each network shall be recorded (see 3.13). Testing shall be as specified in 4.5.1.

TABLE I

| Network and test points | Limits (pf) | Test identification number (see 3.13) |
|---|---|---|
| Nose section, network 1: Tee connector 2CP4 J1 | 464 - 517 | 092001 |
| Nose section, network 2: Tee connector 2CP3 J1 | 467 - 520 | 092003 |
| Overall network 1: 4W70 P3A to P3B-H | 709 - 785 | 092002 |
| Overall network 2: 4W70 P3E to P3F-D | 787 - 879 | 092004 |

3.6 Component continuity loop resistance.- The resistance of each component continuity loop shall not exceed 2.0 ohms when measured between the terminals of the M1103 J2 connector as specified in Table II. In addition, the actual resistance value for each item continuity loop shall be recorded (see 3.13). Testing shall be as specified in 4.5.2.



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TABLE II

| Component | Circuit test points From | To | Test identification number (see 3.13) |
|---|---|---|---|
| M1103 | J2-S | J2-R | 092008 |
| M30 | X | A | 092009 |
| M22 | i | j | 092010 |
| M8A | k | s | 092011 |
| M8B | r | q | 092012 |
| M9A | e | f | 092013 |
| M9B | g | h | 092014 |
| Impact fuze network | U | T | 092015 |
| M4 | W | V | 092016 |
| Warhead continuity circuit | P | N | 092017 |

3.7 M1103 inertial fuze setting.- The inertial fuze shall be capable of accepting the presets as specified in Table III when tested as specified in 4.5.3.

TABLE III. M1103 inertial fuze presets.

| Preset | Synchro readings (degrees ± 5°) Coarse | Fine |
|---|---|---|
| Set decrement | 288 | 101 |
| Set velocity | 216 | 76 |
| Set multiplier | 165 | 223 |
| Set time | 288 | 101 |
| Reset decrement | 0 | 0 |
| Reset velocity | 0 | 0 |
| Reset multiplier | 30 | 41 |
| Reset time | 0 | 0 |

3.8 M22 sensitive switch functioning.- Each of the two M22 sensitive switches shall be functioned (depressed and released) to ensure that it is capable of a make-and-break operation of the warhead section continuity loop circuit. Continuity shall be measured between terminals J8-C and J8-E of the 4W36 wiring harness. Testing shall be as specified in 4.5.4.

3.9 M21A1 rotary switch functioning.- The M21A1 rotary switch shall be functioned to ensure that is will actuate without failure and will cycle correctly from position 0 to 1, 0 to 2, and 0 to 3 (in that sequence). The switch shall be monitored



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for correct cycling to position 0, 1, 2, and 3 by checking continuity between terminals of the 4W36 wiring harness J8 connector which are listed in Table IV. After this test, the switch shall be set to position 0. Testing shall be as specified in 4.5.5.

TABLE IV. M21A1 rotary switch continuity list for positioning.

| Position (see 6.4) | 4W36 wiring harness From | To |
|---|---|---|
| 0 | J8-R | J8-M |
| 1 | R | U |
| 2 | R | f |
| 3 | R | e |

3.10 Warhead section subassembly continuity loop resistance.- The resistance of the warhead section subassembly continuity loop shall be as specified in Table V. In addition, the actual resistance value shall be recorded (see 3.13). Testing shall be as specified in 4.5.6.

TABLE V

| 4W36 wiring harness From | To | Resistance (ohms) | Test identification number (see 3.13) |
|---|---|---|---|
| J8-C | J8-E | 11.0 max. | 092007 |

3.11 Missile power circuit resistance.- The resistance of the missile power (lift-off) networks shall be as specified in Table VI. In addition, the actual resistance value for each network shall be recorded (see 3.13). Testing shall be as specified in 4.5.7.

TABLE VI

| Network and test points | Limits (ohms) | Test identification number (see 3.13) |
|---|---|---|
| Network 1: 4W36 J8-GG to J8-A | 1.9 - 7.0 | 092005 |
| Network 2: 4W36 J8-z to J8-A | 1.9 - 7.0 | 092006 |



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3.12 Weight.- The weight of the completely assembled warhead section subassembly (less the gas generator assembly) shall be as specified in Table VII. In addition, the actual weight in pounds (lb) shall be recorded (see 3.13). Testing shall be as specified in 4.5.8.

TABLE VII

| Limits (lb) | Test identification number (see 3.13) |
|---|---|
| 308.3 - 317.7 | 040001 |

3.13 Serialization.- A six-letter serial code shall be assigned to each serialized production item. The contractor shall obtain these serial codes by means of a letter addressed to: Commander, US Army Armament Research and Development Command, Attn: DRDAR-QAN-R, Dover, New Jersey 07801. The letter of request shall include: (a) exact nomenclature of the item as specified on the applicable drawing (see dwg. 8876120), (b) contract number or procurement request order number, and (c) number of serial codes required (unused codes must be returned). In addition to the normal inspection records, the contractor shall provide data feedback (see 6.1.2) in conformance with Quality Assurance Instruction QAI-PA-1. The data required to be coded, and the test identification numbers, are specified in 3.5, 3.6, 3.10, 3.11, and 3.12.

3.14 Workmanship.- All parts and accessories shall be constructed, assembled, and finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness, marking, cleaning, and freedom of the parts from burrs and sharp edges which may affect performance or result in injury during handling. All parts shall be free of chips, dirt, grease, rust, and other foreign material. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agents.

## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of



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the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.2 Classification of inspections.- The inspection requirements specified herein are classified as follows:

1. First article inspection (see 4.3)
2. Quality conformance inspection (see 4.4)

4.3 First article inspection.

4.3.1 Submission.- When the contract requires first article inspection, the contractor shall submit one warhead section subassembly and one splice ring (dwg. 10595610) to the testing facility designated in the invitation for bids or request for proposal.

4.3.2 Inspection and testing to be performed.- The first article inspection samples shall be inspected, as applicable, for the defects specified in 4.4.2.1.1 and 4.4.2.1.2 using inspection methods contained therein, and for all requirements of Section 3, which are listed in Table VIII, using the test methods specified in 4.5. Inspection and testing shall be performed with the acceptance inspection equipment specified in 4.4.4. The tests listed in Table VIII are in a suggested sequence of testing; however, the tests may be performed in any sequence for convenience purposes. If the subassembly, or splice ring, fails to comply with any of the applicable requirements, the first article samples shall be rejected. The Government reserves the right to terminate its inspections upon any failure of the first article samples to comply with the stated requirements. This series of tests is not considered destructive.

TABLE VIII. First article tests.

| Test | Requirement paragraph |
|---|---|
| 1. Capacitance of the impact fuze networks | 3.5 |
| 2. Component continuity loop resistance | 3.6 |
| 3. M1103 inertial fuze setting | 3.7 |
| 4. M22 sensitive switch functioning | 3.8 |
| 5. M21A1 rotary switch functioning | 3.9 |
| 6. Warhead section subassembly continuity loop resistance | 3.10 |
| 7. Missile power circuit resistance | 3.11 |
| 8. Weight | 3.12 |



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4.4 Quality conformance inspection.

4.4.1 Inspection lot.- A lot or batch and its formation, size, and presentation is described in MIL-STD-105, Section 5. Accordingly, a lot shall mean an inspection lot, and a batch shall mean an inspection batch for the purposes of this specification. The manner in which each inspection lot or batch is to be presented and identified by the contractor shall be designated or approved by the Government representative.

4.4.2 Product inspection examination.- A sample shall be selected at random from each inspection lot in accordance with the major defect or the minor defect table as applicable (see Table IX). Any defect found in the sample, selected for the major or minor classification, shall reject the lot. Critical defects shall require 100 percent inspection of all items in the inspection lot. If any critical defect is found in an item, the item shall be removed from the lot. The sequence of inspections for the classification of defects need not be followed so long as all the defect inspections are performed.

TABLE IX. Inspection lot sampling.

| Major Defect Table | | Minor Defect Table | |
|---|---|---|---|
| Lot Size | Sample Size | Lot Size | Sample Size |
| 1-30 | all | 1-30 | all |
| 31-45 | 29 | 31-45 | 27 |
| 46-60 | 35 | 46-60 | 30 |
| 61-75 | 38 | 61-75 | 33 |
| 76-95 | 42 | 76-95 | 35 |
| 96-130 | 46 | 96-130 | 37 |
| 131-180 | 50 | 131-180 | 39 |
| 181-250 | 52 | 181-250 | 41 |
| 251-400 | 54 | 251-400 | 42 |
| 401 and over | 56 | 401 and over | 44 |



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4.4.2.1 Classification of defects.- Critical, major, and minor defects are defined in MIL-STD-105.

4.4.2.1.1 Ring, Splice (Aft Warhead) (see dwg. 10595610). 1/

| Categories | Defects | Method of Inspection |
|---|---|---|
| Critical: | None defined. | |
| Major: | | |
| 101. | Outside diameter at rear of angular groove | Gage |
| 102. | Location of hole with bushing (2 places) | Gage |
| 103. | Diameter of hole with bushing, min. (2 holes) | Gage |
| 104. | Location of spotface on internal lug (4 places) | Gage |
| 105. | Thickness of lug with spotface, min. (4 lugs) | Gage |
| 106. | Location of cutout (at 5 and at 11 o'clock position)(2 places) | Gage |
| 107. | Length of cutout (at 5 and at 11 o'clock position), min. (2 places) | Gage |
| 108. | Depth of cutout (at 5 and at 11 o'clock position), min. (2 places) | Gage |
| 109. | Depth of angular groove, max | Gage |
| 110. | Angle of angular groove, min | Gage |
| 111. | Flatness of rear surface of ring | Gage |
| Minor: | | |
| 201. | Inside diameter (at angular groove area), max | Gage |
| 202. | Height of splice ring, min | Gage |
| 203. | Configuration of lug improper (6 lugs) | Visual |
| 204. | Protective finish missing or improper | Visual |
| 205. | Evidence of poor workmanship (see 3.14) | Visual |

1/ The splice ring, submitted as part of the first article samples, shall be used for this inspection during first article testing.



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4.4.2.1.2 Warhead Section Subassembly, Atomic: M15 (see dwg. 8876120).

| Categories | Defects | Method of Inspection |
|---|---|---|
| Critical: | None defined. | |
| Major: | | |
| 101. | Torque on spin system bracket mounting screw, min. (3 brackets, 10 screws) | Gage |
| 102. | Torque on nozzle mounting screw, min. (2 nozzles, 8 screws) | Gage |
| 103. | Torque on manifold mounting screw, min. (4 screws) | Gage |
| 104. | Torque on tube assembly sleeve nut, min. (2 nuts) | Gage |
| 105. | Torque on cover plate bolt, min. (12 bolts) | Gage |
| 106. | Torque on bulkhead assembly mounting bolt, min. (36 bolts) | Gage |
| 107. | Mismatch (step up) between sections (2 places)(see dwg. 10561209) | Manual/Gage |
| 108. | Mismatch (step down) between sections, max. (2 places) (see dwg. 10561209) | Manual/Gage |
| 109. | Gap between sections (2 places)(see dwg. 10561209) | Visual/Gage |
| 110. | Imperfections on external surface (including cracks, gouges, irregularities, and voids) | Visual |
| 111. | Any component on bulkhead assembly missing or damaged (excluding gas generator assembly) | Visual |
| Minor: | | |
| 201. | Safety wire (cover plate area) missing or improper (2 places) | Visual |
| 202. | Barrier material (bulkhead area) missing or improper (4 places) | Visual |
| 203. | Bare spots or scratches on external painted surface | Visual |
| 204. | Marking missing, illegible, or improper | Visual |
| 205. | Evidence of poor workmanship (see 3.14) | Visual |



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4.4.2.1.3 Shipping and Storage Container: M483, Packing and Marking, for Warhead Section Subassemblies, Package 1 of 2 (prior to securing cover)(see dwg. 8886530).

| Categories | Defects | Method of Inspection |
|---|---|---|
| Critical: | None defined. | |
| Major: | | |
| 101. | Torque on shear mount bolt, min. (see Packing Instructions notes)..... | Gage |
| 102. | Torque on splice ring nut, min..... | Gage |
| 103. | Torque on support plate nut on rear of suspension frame, min. (2 bolts)..... | Gage |
| 104. | Torque on forward clamp nut, min..... | Gage |
| 105. | Cable assemblies 24W1 and 24W3 improperly assembled and connected..... | Visual |
| 106. | Shear mount improperly located (8 mounts)..... | Visual |
| 107. | Desiccant missing or improper (10 bags)..... | Visual |
| 108. | Bolt and washer (12 each) missing, improper, or improperly attached to desiccant cage..... | Visual |
| Minor: | | |
| 201. | Plastic film, on forward clamp assembly, missing or improper..... | Visual |
| 202. | Protective cover for the 24W1 and the 24W3 cables missing, improper, or improperly packaged (in records receptacle)(2 covers)..... | Visual |
| 203. | Dust cover missing, improper, or improperly attached to unused mounting bracket..... | Visual |
| 204. | Equipment maintenance log form, in records receptacle, missing or improper..... | Visual |

4.4.2.1.4 Shipping and Storage Container: M483, Packing and Marking, for Warhead Section Subassemblies, Package 1 of 2 (see dwg. 8886530).

| Categories | Defects | Method of Inspection |
|---|---|---|
| Critical: | None defined. | |
| Major: | | |
| 101. | Torque on cover tee-bolt nut, min. (28 nuts)..... | Gage |



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| | | |
|---|---|---|
| | 102. | Excessive leakage (see leak test under Packing Instructions notes).....Manual/Gage |
| | 103. | Container damaged (including punctures, dents, or gouges).....Visual |
| Minor: | | |
| | 201. | Scratch on container exposing base metal.....Visual |
| | 202. | Metallic seal missing or improper (8 places).....Visual |
| | 203. | Marking missing, improper, or illegible.....Visual |

4.4.2.1.5 Shipping and Storage Container: M483, Packing and Marking, for Warhead Section Subassemblies, Package 2 of 2 (prior to securing cover)(see dwg. 8886529).

| Categories | Defects | Method of Inspection |
|---|---|---|
| Critical: | None defined. | |
| Major: | | |
| | 101. Gas generator improperly wrapped and bagged..... | Visual/Manual |
| | 102. Cushioning disk (2 places) and cushioning material improperly assembled..... | Visual/Manual |
| Minor: | | |
| | 201. Bag marking missing, improper, or illegible..... | Visual |

4.4.2.1.6 Shipping and Storage Container: M483, Packing and Marking, for Warhead Section Subassemblies, Package 2 of 2 (see dwg. 8886529).

| Categories | Defects | Method of Inspection |
|---|---|---|
| Critical: | None defined. | |
| Major: | | |
| | 101. Torque on cover sealing nut, min. (see dwg. 8806652)..... | Gage |
| | 102. Drum improperly sealed (including metallic seal)(see dwg. 8806652)..... | Visual |
| | 103. Drum damaged (including punctures, dents, or gouges)..... | Visual |



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Minor:

201. Scratch on drum exposing base metal.....Visual

202. Marking missing, improper, or illegible.....Visual

4.4.3 Inspection testing.- The contractor shall perform all of the tests specified in Table X. The tests listed in Table X are in a suggested sequence of testing; however, the tests may be performed in any sequence for convenience purposes. None of these tests is considered destructive, and samples so tested shall be returned to the lot.

TABLE X. Quality conformance tests.

| Test | Test classification |
|---|---|
| 1. Capacitance of the impact fuze networks | Major |
| 2. Component continuity loop resistance | Major |
| 3. M1103 inertial fuze setting | Major |
| 4. M22 sensitive switch functioning | Major |
| 5. M21A1 rotary switch functioning | Major |
| 6. Warhead section subassembly continuity loop resistance | Major |
| 7. Missile power circuit resistance | Major |
| 8. Weight | Major |

4.4.3.1 Capacitance of the impact fuze networks (see 3.5), component continuity loop resistance (see 3.6), M1103 inertial fuze setting (see 3.7), M22 sensitive switch functioning (see 3.8), M21A1 rotary switch functioning (see 3.9), warhead section subassembly continuity loop resistance (see 3.10), missile power circuit resistance (see 3.11), and weight (see 3.12).- Each warhead section subassembly shall be subjected to these tests (see applicable paragraph in 4.5). If any subassembly fails to comply with the applicable requirements, it shall be classified defective and be removed from the lot.



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4.4.4 Acceptance inspection equipment.

4.4.4.1 Major defects and tests.- Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration procedures designed or specified by the contractor (see 6.1.2). The contractor shall obtain approval of such equipment designs, operating instructions, and calibration procedures prior to use on the contract (see 6.3).

4.4.4.2 Minor defects and tests.- The acceptance inspection equipment, operating instructions, and calibration procedures used by the contractor for minor classification inspection shall be approved by the Government representative responsible for acceptance inspection.

4.5 Methods of inspection.- The tolerances specified in this specification are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used. During testing, the warhead section subassembly shall be placed on a suitable handling stand, and the center section, aft section, and nose section shall be separated, as applicable, to facilitate testing. Unless otherwise specified, the tests shall be conducted at room temperature (77° ± 10°F).

4.5.1 Capacitance of the impact fuze networks.- The capacitance of the impact fuze networks shall be measured by using a Government-approved capacitance bridge at a frequency of 1000 hertz (Hz). The nose section shall be fastened to the center section prior to performing the tests on the overall networks. The impact fuze networks shall comply with the requirements of 3.5 during this test.

4.5.2 Component continuity loop resistance.- The resistance of each component continuity loop shall be measured with a Government-approved resistance measuring device. Before doing this test, perform the following operations: (a) depress both M22 switches, (b) connect a jumper wire across terminals D and E of the 4W67 wiring harness P2 connector, and (c) connect a jumper wire across terminals C and E of the M1103 inertial fuze P8 connector. The components shall comply with the requirements of 3.6 during this test.



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4.5.3 M1103 inertial fuze setting.- A Government-approved fuze setter shall be used to apply the presets specified in Table III through the 4W36 J8 connector (located on the aft bulkhead of the M15 subassembly). The synchro transmitters, in the fuze setter, shall be used to verify the presets. The presetting circuits, through the 4W36 J8 connector, and the applicable voltage inputs are specified in Table XI. The fuze shall comply with the requirements of 3.7 during this test.

TABLE XI. M1103 fuze setting inputs.

| Signal | 4W36 cable From | 4W36 cable To | Voltage |
|---|---|---|---|
| Brake excitation: | | | |
| (a) Decrement | J8-y | J8-A | 28 ± 3.5 VDC |
| (b) Velocity | k | A | " |
| (c) Multiplier | MM | A | " |
| (d) Time | r | A | " |
| Motor control phase: | | | |
| (a) Decrement | KK | JJ | 2.5 Vrms max., 400 ± 4 Hz |
| (b) Velocity | j | i | 3.5 " |
| (c) Multiplier | q | p | 5.5 " |
| (d) Time | t | s | 4.5 " |
| Motor reference phase | AA | T | 26 ± 1.5 Vrms, 400 ± 4 Hz |
| Synchro rotor excitation | T | L | " |

4.5.4 M22 sensitive switch functioning.- The functioning of the M22 sensitive switches, as evidenced by monitoring continuity of the warhead section continuity loop circuit, shall be determined by means of a Government-approved continuity tester. Before doing this test, perform the following operations: (a) connect a jumper wire across terminals P2-D and P2-E of the 4W67 wiring harness and (b) connect a jumper wire across terminals P8-C and P8-E of the M1103 inertial fuze. Connect the continuity tester between terminals J8-C and J8-E of the 4W36 wiring harness. The continuity tester shall indicate continuity when both M22 switches are depressed and an open circuit when either one of the M22 switches is released. Each switch shall be functioned separately to ensure that it will open the warhead section continuity loop circuit. The switches shall comply with the requirements of 3.8 during this test.



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4.5.5 M21A1 rotary switch functioning.- The M21A1 rotary switch (initially at position 0) shall be functioned to positions 1, 0, 2, 0, 3, and 0 (in that sequence). While the switch is at each position, a Government-approved continuity tester shall be used to ensure that the switch is correctly positioned by checking continuity between the terminals listed in Table IV. The switch shall comply with the requirements of 3.9 during this test.

4.5.6 Warhead section subassembly continuity loop resistance.- The resistance of the warhead section subassembly continuity loop shall be measured by using a Government-approved resistance measuring device. Before doing this test, perform the following operations: (a) depress both M22 switches, (b) connect a jumper wire across terminals P2-D and P2-E of the 4W67 wiring harness, and (c) connect a jumper wire across terminals P8-C and P8-E of the M1103 inertial fuze. The subassembly shall comply with the requirements of 3.10 during this test.

4.5.7 Missile power circuit resistance.- The resistance of the missile power circuit shall be measured by using a Government-approved resistance measuring device between the test points listed in Table VI. The missile power circuit shall comply with the requirements of 3.11 during this test.

4.5.8 Weight.- The warhead section subassembly shall be weighed by using a Government-approved weighing device and shall comply with the requirements of 3.12.

## 5. PACKAGING

5.1 Preservation, packaging, packing, and marking.- Preservation, packaging, packing, and marking of the gas generator and of the warhead section subassembly shall be in accordance with dwg. 8886529 and 8886530 respectively.

## 6. NOTES

6.1 Ordering data.- Procurement documents should specify the following:

6.1.1 Procurement requirements.- Title, number, and date of this specification and first article sample requirements.

6.1.2 Data requirements.- Data specified in 3.5, 3.6, 3.10, 3.11, and 3.12, and contractor engineering documentation specified in 4.4.4.1, will be delivered as identified on a numbered DD Form 1664 when specified on a DD Form 1423 incorporated into the contract.



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6.2 Objective evidence.- Records of the contractor's quality control and inspections which can be verified.

6.3 Acceptance inspection equipment.- The contractor shall obtain approval of equipment designs, operating instructions, and calibration procedures from Commander, US Army Armament Research and Development Command, Attn: DRDAR-QAN-I, Dover, New Jersey 07801. NOTE: If the assembly and packaging of the warhead section subassembly is to be performed at ARRADCOM, Dover, New Jersey, the inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration procedures as specified in the Equipment Lists listed in the Equipment Tabulation, ET-8876120.

6.4 M21A1 rotary switch positioning.- To set the M21A1 rotary switch to the desired position, apply 28 ± 4 VDC between the J8 connector terminals of the 4W36 wiring harness which are listed in Table XII. Polarity shall be as specified in the table.

TABLE XII

| Position | 4W36 wiring harness | |
|---|---|---|
| | From (+) | To (-) |
| 0 | J8-Z | J8-d |
| 1 | W | d |
| 2 | G | d |
| 3 | V | d |

6.5 Drawings and publications.- Drawings and publications listed in Section 2 of this specification under the heading of US Army Armament Research and Development Command (ARRADCOM) may also include drawings and publications prepared by, and identified as, Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, or Picatinny Arsenal drawings and publications. Technical data originally prepared by these activities is now under the cognizance of ARRADCOM.



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6.6 Changes from previous issue.- Asterisks are not used in this revision, to identify changes with respect to the previous issue, due to the extensiveness of the changes.

6.7 Supersession data.- This specification includes the requirements of Purchase Description ABMA-PD-W-858(Rev. 4), dated 13 December 1965, with Amendment 3 dated 6 June 1974.

Custodian:
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